Solving 1D & 2D Helmholtz Equation using Finite Element Analysis

Mano
`mano@novapropulsion.space`
`github.com/Samson-Mano`

11 October 2025

Abstract

This article details the **Finite Element Analysis (FEA)** approach for solving the **one-dimensional (1D) and two-dimensional (2D) Helmholtz equation**, which arises from the time-harmonic reduction of Maxwell's equations. The work establishes a clear framework by first examining the impact of **various boundary conditions**, including Dirichlet, Neumann, Sommerfeld, and Absorbing Boundary Conditions (ABC). Following this, the complete methodology for the **weak form finite element formulation** is derived. The discretization process, utilizing **linear finite elements**, is shown, culminating in the explicit steps for the **assembly of the global system matrices**. The numerical solution is implemented in **C#, openTK & C++**, and the resulting wave fields for all discussed boundary conditions are analyzed, providing a clear comparison of their effects on the modelled wave propagation. This study validates the use of FEA as a robust and accurate method for tackling boundary-value problems in wave mechanics.

Contents

1 Introduction

The Helmholtz equation is the time-harmonic form of the wave equation and appears in acoustics, electromagnetics and quantum mechanics. In its scalar 1D form it is typically written as

$$\frac{d^2u}{dx^2} + k^2 u = -f(x) \quad \text{in } \Omega, \tag{1}$$

where $u(x)$ is the complex amplitude (phasor), k is the wavenumber ($k = \omega/c$) for waves in a medium with wave speed (c), and $f(x)$ is a source term.

1.1 Short historical note

The equation is named after Hermann von Helmholtz (1821–1894). It became central to mathematical physics in the 19th century while studying vibrations, acoustics and later electromagnetic waves.

1.2 Derivation from Maxwell's equations (Time-Harmonic Fields)

We begin the derivation of the Helmholtz equation from Maxwell's curl equations in the time domain, assuming linear, isotropic, and homogeneous media ($\mathbf{D} = \varepsilon\mathbf{E}$ and $\mathbf{B} = \mu\mathbf{H}$):

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \qquad \nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \tag{2}$$

1.2.1 Frequency Domain Transformation

Assuming **time-harmonic fields** with a time dependence of $e^{i\omega t}$ (where ω is the angular frequency), the time derivatives transform into multiplication by $i\omega$ ($\partial/\partial t \to i\omega$). The equations in the frequency domain become:

$$\nabla \times \mathbf{E} = -i\omega\mu\mathbf{H} \tag{3}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + i\omega\varepsilon\mathbf{E} \tag{4}$$

1.2.2 Deriving the Vector Helmholtz Equation

To obtain an equation in terms of **E** only, we take the curl of Equation (3) and substitute Equation (4) into the result:

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla \times (-i\omega\mu\mathbf{H}) = -i\omega\mu(\nabla \times \mathbf{H})$$

Substituting Equation (4):

$$\nabla \times (\nabla \times \mathbf{E}) = -i\omega\mu(\mathbf{J} + i\omega\varepsilon\mathbf{E})$$

Next, we employ the vector identity $\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2\mathbf{E}$:

$$\nabla(\nabla \cdot \mathbf{E}) - \nabla^2\mathbf{E} = -i\omega\mu\mathbf{J} + \omega^2\mu\varepsilon\mathbf{E}$$

Rearranging terms and defining the **wavenumber** $k^2 = \omega^2\mu\varepsilon$, we arrive at the **vector Helmholtz equation**:

$$\nabla^2\mathbf{E} + k^2\mathbf{E} = \nabla(\nabla \cdot \mathbf{E}) + i\omega\mu\mathbf{J} \tag{5}$$

1.2.3 Reduction to 1D Scalar Form

For the **homogeneous, source-free case** in a simple medium (where $\mathbf{J} = \mathbf{0}$ and $\nabla \cdot \mathbf{E} = 0$), the right-hand side of Equation (5) is zero. This yields the source-free 1D scalar Helmholtz equation:

$$\frac{d^2u}{dx^2} + k^2u = 0. \tag{6}$$

However, in many practical applications, we consider a non-homogeneous domain or excitation. By considering only wave propagation along a single spatial dimension (x) and a single field component (u), we focus on the general **1D scalar Helmholtz boundary value problem** on the domain $\Omega = (0, L)$:

$$\frac{d^2u}{dx^2} + k^2u = -f(x), \qquad x \in (0, L), \tag{7}$$

where $f(x)$ represents the known **source term** (e.g., an external excitation or an equivalent force term).

The problem is completed by applying boundary conditions (BCs). For example, a simple Dirichlet condition is $u(0) = u(L) = 0$. Variations such as Neumann, Robin, and radiating (Sommerfeld/Absorbing) BCs are crucial for modeling wave reflection and open-domain scattering problems.

2 Conditions for the 1D Helmholtz Equation

Boundary conditions (BCs) are essential constraints applied at the domain boundaries that, when coupled with the differential equation, yield a unique solution. In the context of the Finite Element Method (FEM), these conditions are generally classified based on how they enter the weak form of the problem:

2.1 Standard Boundary Conditions

- **Dirichlet (Essential) BC** The Dirichlet boundary condition, also known as the **essential** or **geometric** boundary condition, prescribes the value of the primary field variable (the state variable) directly on the boundary of the domain. For the 1D Helmholtz equation, this specifies the field value u at the endpoints, e.g., $u(0) = u_0$ or $u(L) = u_L$. This condition is imposed directly on the function space.

- **Neumann (Natural) BC** The Neumann boundary condition, or **natural** boundary condition, prescribes the value of the **normal derivative** (or flux/gradient) of the state variable on the boundary. For the 1D problem, this is often written as $\frac{du}{dx}\big|_{x=0} = g_0$. Physically, this describes the injection or removal of flux/energy at the boundary. This condition naturally arises from the integration by parts in the weak formulation.

- **Robin (Mixed) BC** The Robin boundary condition, or **mixed** boundary condition, is a linear combination of the Dirichlet and Neumann conditions. It prescribes a relationship between the field variable u and its normal derivative $\frac{du}{dn}$ on the boundary, e.g., $\frac{du}{dx} + \alpha u = \beta$. This condition is crucial for modeling imperfectly reflecting or impedance-matched surfaces.

2.2 Radiation and Absorbing Boundary Conditions (ABCs)

For computational problems involving wave scattering in infinite or semi-infinite domains, the simulation must be truncated at a finite distance, $x = L$. The boundary condition imposed at this artificial outer surface must prevent non-physical reflections and mimic wave propagation into the far-field.

- **Sommerfeld Radiation Condition** The Sommerfeld radiation condition is the analytically necessary condition at **x** $\to \infty$ to ensure the uniqueness of the solution in an unbounded domain. Physically, it dictates that only outward-traveling waves are allowed, preventing scattered energy from entering the domain from infinity. For the 1D Helmholtz equation, the condition for $x \to \infty$ simplifies to require the scattered wave u_s to satisfy:

$$\lim_{x\to\infty}\left(\frac{du_s}{dx} - iku_s\right) = 0$$

where k is the wavenumber.

- **Absorbing Boundary Conditions (ABCs)** Since the Sommerfeld condition is a limit process that cannot be enforced at a finite boundary $x = L$, it must be replaced by an **Absorbing Boundary Condition (ABC)** at the artificial truncation boundary. The goal of an ABC is to absorb outgoing waves without reflection. The accuracy of the interior solution is highly dependent on the accuracy of the ABC used; more precise interior discretizations demand more sophisticated and accurate ABCs to maintain overall solution quality. Common examples include the first-order ABC (which perfectly absorbs normally incident waves) or more complex Perfectly Matched Layers (PMLs).

3 Finite Element Formulation and Discretization

This section details the transformation of the governing differential equation into its weak form and the subsequent Finite Element Method (FEM) discretization using the Galerkin weighted residual approach.

3.1 The Weak Form Derivation

We begin with the general **1D scalar Helmholtz boundary value problem** on the domain $\Omega = (0, L)$:

$$\frac{d^2u}{dx^2} + k^2 u = -f(x), \qquad x \in (0, L), \tag{8}$$

where $u(x)$ is the unknown field variable, k is the wavenumber, and $f(x)$ is the known **source term**.

The **Galerkin weighted residual method** requires multiplying the strong form by a smooth test function $v(x)$ and integrating over the domain Ω:

$$\int_\Omega \left(\frac{d^2u}{dx^2} + k^2 u \right) v \, dx = - \int_\Omega f v \, dx.$$

To obtain the **weak form**, we apply **integration by parts** to the second-order term. This lowers the continuity requirement on the trial function u and naturally introduces the boundary conditions:

$$\int_\Omega \frac{d^2u}{dx^2} v \, dx = \left[\frac{du}{dx} v \right]_{\partial\Omega} - \int_\Omega \frac{du}{dx} \frac{dv}{dx} \, dx$$

Substituting this back and rearranging yields the final **weak form**: Find $u \in V$ such that for all test functions $v \in V$:

$$\int_\Omega \left(\frac{du}{dx} \frac{dv}{dx} - k^2 uv \right) \, dx = \int_\Omega f v \, dx - \left[\frac{du}{dx} v \right]_{\partial\Omega} \tag{9}$$

The term $\left[\frac{du}{dx} v\right]_{\partial\Omega}$ is the **boundary term** where Neumann and Robin type boundary conditions will be imposed.

3.2 Finite Element Discretization

In the FEM approach, the domain Ω is divided into a finite number of elements. Within each element, the unknown solution $u(x)$ is approximated by a linear combination of nodal values u_j and local **shape functions** $N_j(x)$:

$$u(x) \approx u^h(x) = \sum_{j=1}^{n} u_j N_j(x)$$

The shape functions $N_j(x)$ are piecewise polynomial functions that provide the interpolation between nodes. These functions are illustrated in Figure 1.

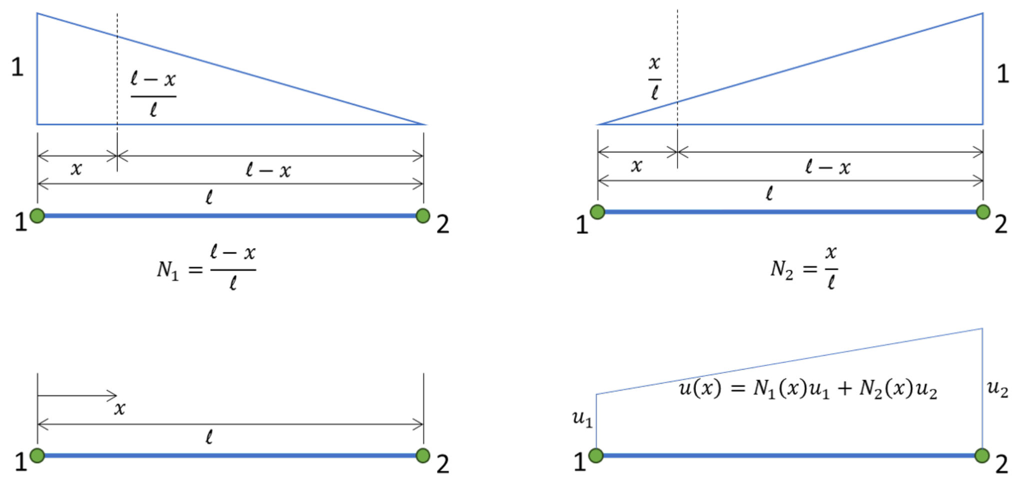

Figure 1: 1D Shape functions

By choosing the test functions $v(x)$ to be the same as the shape functions (the essence of the Galerkin method), $v_i = N_i$, and substituting the approximation u^h into the weak form (9), we transform the continuous problem into a system of algebraic equations:

For a single element $\Omega_e = [x_a, x_b]$ with length $h = x_b - x_a$, we utilize two-node linear shape functions $N_1(x)$ and $N_2(x)$.

The approximations for the field $u^e(x)$ and test function $v^e(x)$ are:

$$u^e(x) = u_1 N_1(x) + u_2 N_2(x) = \mathbf{N}(x)\mathbf{u}^e$$

$$v^e(x) = v_1 N_1(x) + v_2 N_2(x) = \mathbf{N}(x)\mathbf{v}^e$$

The linear shape functions and their derivatives are:

Node	Shape Function $N_j(x)$	Derivative $N_j'(x)$
1	$N_1(x) = \frac{x_b - x}{h}$	$N_1'(x) = -\frac{1}{h}$
2	$N_2(x) = \frac{x - x_a}{h}$	$N_2'(x) = \frac{1}{h}$

The weak form integral on a single element Ω_e is discretized into the elemental system:

$$\mathbf{v}^T \left(\mathbf{K}^e - k^2 \mathbf{M}^e \right) \mathbf{u}^e = \mathbf{v}^T \mathbf{F}^e$$

3.2.1 Element Stiffness (Conduction) Matrix, $\mathbf{K}^e$

The stiffness matrix components are given by $\mathbf{K}_{ij}^e = \int_{\Omega_e} N_i'(x) N_j'(x)\, dx$. Substituting the derivatives:

$$\mathbf{K}^e = \int_{x_a}^{x_b} \frac{1}{h^2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} dx = \frac{1}{h^2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \int_{x_a}^{x_b} dx$$

Since $\int_{x_a}^{x_b} dx = h$, the final matrix is:

$$\mathbf{K}^e = \frac{1}{h} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \tag{10}$$

3.2.2 Element Mass Matrix, $\mathbf{M}^e$

The mass matrix arises from the reaction term in the weak form, $\mathbf{M}_{ij}^e = \int_{\Omega_e} N_i(x) N_j(x)\, dx$. Utilizing the standard integral results for linear basis functions:

$$\mathbf{M}^e = \frac{h}{6} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \tag{11}$$

3.2.3 Element Load Vector, $\mathbf{F}^e$

The element load vector comes from the source term integral: $\mathbf{F}_i^e = \int_{\Omega_e} f(x) N_i(x)\, dx$.

Assuming the source term $f(x)$ is approximately constant over the element, $f(x) \approx f_e$:

$$\mathbf{F}^e = f_e \int_{x_a}^{x_b} \begin{pmatrix} N_1(x) \\ N_2(x) \end{pmatrix} dx$$

Since the area under each linear shape function is $h/2$:

$$\mathbf{F}^e = f_e \frac{h}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{12}$$

3.2.4 Final Element System

The elemental system of equations for the 1D Helmholtz problem is:

$$\left(\mathbf{K}^e - k^2 \mathbf{M}^e\right) \mathbf{u}^e = \mathbf{F}^e$$

Substituting the derived matrices:

$$\left[\frac{1}{h} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} - k^2 \frac{h}{6} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}\right] \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = f_e \frac{h}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{13}$$

4 Assembly of Global System Matrices and Boundary Conditions

The final step in the Finite Element Method (FEM) is to assemble the elemental contributions into a global system and apply the necessary boundary conditions to solve for the unknown nodal field values **u**.

4.1 Global Assembly

The global system matrix **A** and the global load vector **B** are formed by summing the contributions from all N_{el} elements over the domain Ω:

$$\mathbf{A} = \sum_{e=1}^{N_{el}} (\mathbf{K}^e - k^2 \mathbf{M}^e)$$

$$\mathbf{B} = \sum_{e=1}^{N_{el}} \mathbf{F}^e + \mathbf{B}_{\partial\Omega}$$

The final discretized system takes the form:

$$\mathbf{A}\mathbf{u} = \mathbf{B} \tag{14}$$

where $\mathbf{B}_{\partial\Omega}$ is the vector representing boundary fluxes (Neumann, Sommerfeld) applied at the nodes.

4.2 Implementation of Boundary Conditions

The implementation method depends on the type of boundary condition, as each affects the final linear system, Equation (14), differently.

4.2.1 Dirichlet (Essential) Boundary Condition

The Dirichlet condition prescribes the value of the solution at a specific node, $u_i = u_0$. The preferred implementation method is the **elimination method** (or substitution method), which modifies the system to enforce the known value.

If node i has a prescribed value $u_i = u_0$, the global system is modified:

- **Load Vector Modification:** The contribution of the known term is moved to the right-hand side. For all other equations j, the term $A_{j,i}u_0$ is calculated and subtracted from B_j.

- **Matrix Modification:** Row i is replaced to enforce $u_i = u_0$:
  - Set the diagonal entry: $A_{i,i} = 1$.
  - Set all off-diagonal entries in row i and column i to zero ($A_{i,j} = 0$ and $A_{j,i} = 0$ for $i \neq j$).
  - Set the load vector entry: $B_i = u_0$.

The operation on the load vector conceptually follows:

$$\mathbf{B}_{\text{new}} = \mathbf{B}_{\text{old}} - \mathbf{Au}_{\text{known}}$$

4.2.2 Neumann (Natural) Boundary Condition

The Neumann condition is a **natural boundary condition** because it prescribes the flux (gradient) at the boundary, $\frac{du}{dx}\Big|_{\partial\Omega} = q$. This condition is incorporated **naturally** through the boundary term arising from the integration by parts in the weak formulation:

$$\int_{\Omega} \left(\frac{du}{dx}\frac{dv}{dx} - k^2 uv \right) \, dx = \int_{\Omega} f v \, dx - \left[\frac{du}{dx} v \right]_{\partial\Omega}$$

The boundary term, $\left[\frac{du}{dx}v\right]_{\partial\Omega}$, must be evaluated at the domain endpoints. For the 1D domain $\Omega = (0, L)$, the term expands to:

$$\left[\frac{du}{dx} v \right]_0^L = \frac{du}{dx}(L)\, v(L) - \frac{du}{dx}(0)\, v(0). \tag{15}$$

Where the prescribed flux $\frac{du}{dx}$ is known (Neumann condition), the term does not vanish and contributes directly to the global load vector **B**.

Incorporation into the Global Load Vector In the final assembled system $\mathbf{Au} = \mathbf{B}$, the components of the global load vector are: $\mathbf{B} = \mathbf{F}^{\text{internal}} - \mathbf{B}_{\partial\Omega}$. Since the test function v is equivalent to the global basis function N_i, $v(L) = N_N(L) = 1$ and $v(0) = N_1(0) = 1$.

Let $q(0)$ be the prescribed flux at $x = 0$ (Node 1) and $q(L)$ be the prescribed flux at $x = L$ (Node N, the last node). Substituting q for $\frac{du}{dx}$ in Equation (15) and applying the necessary sign changes to move the term to the right-hand side, the contributions to the global load vector **B** are:

- **At $x = L$ (Node N):** The term $\frac{du}{dx}(L)\, v(L)$ contributes $+q(L) \cdot 1$ to the load vector entry B_N.

- **At $x = 0$ (Node 1):** The term $-\frac{du}{dx}(0)\, v(0)$ contributes $-q(0) \cdot 1$ to the load vector entry B_1.

The known flux values are therefore **added directly** (with the appropriate sign determined by the weak form) to the corresponding entries in the global load vector **B**:

$$B_N = B_N^{\text{internal}} + q(L)$$
$$B_1 = B_1^{\text{internal}} - q(0)$$

This process accurately incorporates the physical flux (force) into the system's external load vector.

4.2.3 Sommerfeld (Absorbing) Boundary Condition (ABC)

The **Sommerfeld radiation condition** is used to simulate an open or infinite domain by absorbing outgoing waves at the boundary, thereby preventing artificial reflections. For a 1D Helmholtz problem, the first-order Sommerfeld condition is expressed as:

$$\frac{du}{dx} - iku = 0 \quad \text{at } x = L$$

where k is the wavenumber and u is the wave field.

This is a **Robin (mixed) boundary condition**, since it couples both the function value u and its derivative $\frac{du}{dx}$ on the boundary. Substituting this condition into the weak form ensures that the artificial boundary behaves as an **absorbing layer** rather than a hard reflector.

Derivation from the Weak Form Starting from the weak form of the Helmholtz equation:

$$\int_\Omega \left(\frac{du}{dx}\frac{dv}{dx} - k^2 uv \right) dx = \int_\Omega f v \, dx - \left[\frac{du}{dx} v \right]_{\partial\Omega}$$

At the boundary $x = L$, the Sommerfeld condition replaces the derivative term:

$$\left. \frac{du}{dx} \right|_{x=L} = iku(L)$$

Substituting this into the boundary term yields:

$$-\left[\frac{du}{dx} v \right]_{x=L} = -ik\, u(L)\, v(L)$$

Hence, the weak form becomes:

$$\int_\Omega \left(\frac{du}{dx}\frac{dv}{dx} - k^2 uv \right) \, dx - ik \, u(L) \, v(L) = \int_\Omega f v \, dx$$

The last term represents the additional boundary contribution due to the absorbing condition.

Incorporation into the Global System In matrix form, this results in an additional contribution to the global stiffness matrix **A**:

$$\mathbf{A}\mathbf{u} = \mathbf{B}, \quad \text{with} \quad \mathbf{A} = \mathbf{A}^{\text{internal}} + \mathbf{A}^{\text{ABC}}$$

where the Sommerfeld boundary contribution is:

$$\mathbf{A}^{\text{ABC}} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & (-ik) \end{bmatrix}$$

That is, only the **diagonal entry corresponding to the boundary node** (here node N) is modified:

$$A_{N,N} = A_{N,N}^{\text{internal}} - ik$$

This term behaves like a complex impedance, which absorbs the outgoing energy and minimizes reflection.

Implementation in 1D FEM For a 1D element mesh:

- The Sommerfeld ABC is applied only at the **boundary node** N (the end of the domain).
- In practice, this is implemented numerically as:

  $$A[N, N] \mathrel{+}= -ik$$

  if the global matrix **A** is complex-valued.

This small modification ensures that the finite element solution behaves as if the domain extends infinitely beyond $x = L$.

Extension to 2D FEM For a 2D Helmholtz problem, the Sommerfeld (absorbing) condition applies along the artificial boundaries Γ_{ABC} (typically the outer edges of the computational domain). The weak form includes the boundary integral:

$$\int_{\Gamma_{\text{ABC}}} \left(\frac{\partial u}{\partial n} - iku \right) v \, d\Gamma = 0$$

Substituting $\frac{\partial u}{\partial n} = iku$ gives an additional boundary term:

$$-ik \int_{\Gamma_{\text{ABC}}} uv \, d\Gamma$$

In the finite element assembly process, this leads to an extra matrix contribution for each boundary element:

$$\mathbf{A}^{(e)}_{\text{ABC}} = -ik \int_{\Gamma_e} \mathbf{N}^T \mathbf{N} \, d\Gamma$$

which is added to the corresponding rows and columns of the global stiffness matrix **A**.

Thus, in 2D:

- The ABC modifies the **local edge matrices** along the absorbing boundaries.
- The modified entries are assembled into the global matrix in the same way as the regular element stiffness matrices.

Physical Interpretation The $-ik$ term acts as a complex-valued impedance that allows outgoing waves to pass through the boundary without reflection. Higher-order absorbing boundary conditions or Perfectly Matched Layers (PMLs) extend this concept for more accurate absorption at oblique angles.

Dimension	**Boundary Location**	**Modification**	**Global Effect**
1D	Node N	$A_{NN} \mathrel{+}= -ik$	Adds complex impedance at en
2D	Boundary edges Γ_{ABC}	$\mathbf{A}^{(e)}_{\text{ABC}} = -ik \int_{\Gamma_e} N^T N \, d\Gamma$	Absorbs outgoing wave energy

Summary of Implementation This treatment ensures that the Sommerfeld (absorbing) boundary accurately simulates open-space wave propagation while keeping the finite computational domain finite.

5 Two-Dimensional Finite Element Formulation

This section extends the approach from the 1D problem to the **two-dimensional (2D) scalar Helmholtz boundary value problem** defined on a domain $\Omega \subset \mathbb{R}^2$.

The strong form of the 2D Helmholtz equation is:

$$\nabla^2 u + k^2 u = -f(x, y) \quad \text{in } \Omega \tag{16}$$

where $\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$, $u(x, y)$ is the unknown field variable, k is the wavenumber, and $f(x, y)$ is the known source term.

5.1 Weak Form Derivation in 2D

We apply the Galerkin weighted residual method by multiplying the strong form by a test function $v(x, y)$ and integrating over the domain Ω:

$$\int_\Omega (\nabla^2 u + k^2 u) v \, dA = -\int_\Omega f v \, dA$$

To reduce the continuity requirement on u and to incorporate the natural boundary conditions, we apply **Green's First Identity** (the 2D analogue of integration by parts) to the Laplacian term $\int_\Omega (\nabla^2 u) v \, dA$. Green's First Identity states:

$$\int_\Omega (\nabla^2 u) v \, dA = -\int_\Omega \nabla u \cdot \nabla v \, dA + \oint_{\partial\Omega} v \, (\nabla u \cdot \mathbf{n}) \, d\Gamma \tag{17}$$

In this identity:

- The dot product of the gradients is $\nabla u \cdot \nabla v = \frac{\partial u}{\partial x}\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\frac{\partial v}{\partial y}$.
- The term $\nabla u \cdot \mathbf{n} = \frac{\partial u}{\partial n}$ is the normal derivative of u (the flux) on the boundary $\partial\Omega$.

5.2 The Final Weak Form

Substituting Green's First Identity (17) back into the residual equation yields the final **weak form** of the 2D Helmholtz equation: Find $u \in V$ such that for all test functions $v \in V$:

$$\int_\Omega \left(\nabla u \cdot \nabla v - k^2 u v \right) \, dA = \int_\Omega f v \, dA + \oint_{\partial\Omega} v \left(\frac{\partial u}{\partial n} \right) \, d\Gamma \tag{18}$$

The boundary integral term, $\oint_{\partial\Omega} v \left(\frac{\partial u}{\partial n}\right) \, d\Gamma$, is the mechanism through which the Neumann (prescribed flux) and Sommerfeld (Absorbing) boundary conditions are imposed.

5.3 Finite Element Discretization using Triangular Elements

In the FEM approach, the 2D domain Ω is discretized into a finite number of three-noded triangular elements. The unknown solution $u(x, y)$ is approximated within each element using the nodal values u_i and the local **shape functions** $N_i(x, y)$:

$$u(x,y) \approx u^e(x,y) = \sum_{i=1}^{3} u_i N_i(x,y) = \mathbf{N}(x,y)\mathbf{u}^e \tag{19}$$

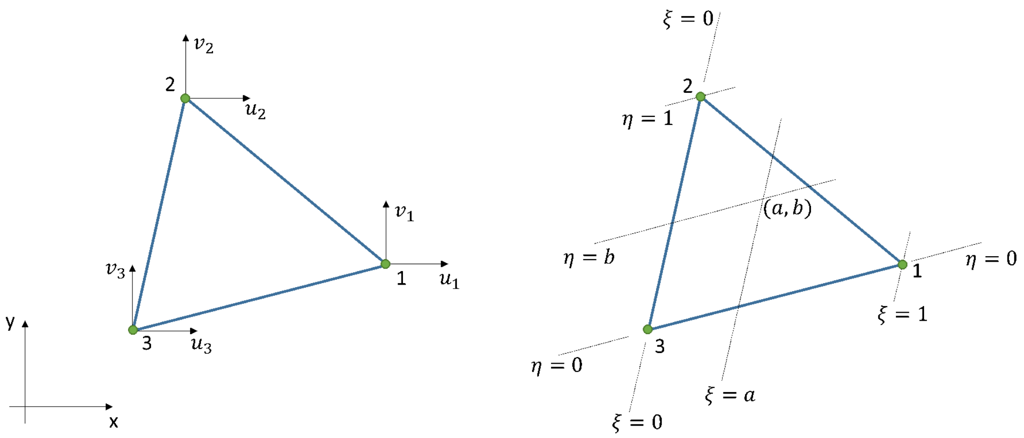

Figure 2: Linear Triangular element

5.3.1 Linear Triangular Element Shape Functions

The shape function $N_i(x, y)$ for the 3-noded linear element is a linear polynomial satisfying the **Kronecker Delta property**: $N_i(x_j, y_j) = \delta_{ij}$.

The global Cartesian shape functions are defined as:

$$N_i = \frac{1}{2A_e}(a_i + b_i x + c_i y)$$

where A_e is the area of the element, and the coefficients a_i, b_i, c_i are defined by the nodal coordinates (x_i, y_i), (x_j, y_j), and (x_k, y_k):

$$\begin{aligned}
a_i &= x_j y_k - x_k y_j, & b_i &= y_j - y_k, & c_i &= x_k - x_j \\
a_j &= x_k y_i - x_i y_k, & b_j &= y_k - y_i, & c_j &= x_i - x_k \\
a_k &= x_i y_j - x_j y_i, & b_k &= y_i - y_j, & c_k &= x_j - x_i
\end{aligned}$$

5.3.2 Gradient Operator Matrix, B

The required derivatives are packaged into the **B** matrix, which relates the field gradient ∇u^e to the nodal values $\mathbf{u}^e$:

$$\nabla u^e = \begin{pmatrix} \partial u^e/\partial x \\ \partial u^e/\partial y \end{pmatrix} = \mathbf{B}\mathbf{u}^e$$

Since the derivatives $\frac{\partial N_i}{\partial x} = \frac{b_i}{2A_e}$ and $\frac{\partial N_i}{\partial y} = \frac{c_i}{2A_e}$ are constant, the **B** matrix is:

$$\mathbf{B} = \frac{1}{2A_e} \begin{pmatrix} b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix} \tag{20}$$

5.3.3 Element Matrices for the Helmholtz Equation

The elemental system $(\mathbf{K}^e - k^2\mathbf{M}^e)\,\mathbf{u}^e = \mathbf{F}^e$ is obtained from the weak form, Equation (18).

A. Element Stiffness (Conduction) Matrix, $\mathbf{K}^e$ This matrix arises from the gradient term $\int_{\Omega_e} \nabla u \cdot \nabla v \, dA = \int_{\Omega_e} (\mathbf{B}\mathbf{u}^e)^T(\mathbf{B}\mathbf{v}^e)\, dA$. Since **B** is constant:

$$\mathbf{K}^e = \int_{\Omega_e} \mathbf{B}^T\mathbf{B}\, dA = A_e\, \mathbf{B}^T\mathbf{B} \tag{21}$$

B. Element Mass Matrix, $\mathbf{M}^e$ This matrix arises from the reaction term $\int_{\Omega_e} uv\, dA = \int_{\Omega_e} (\mathbf{N}\mathbf{u}^e)^T(\mathbf{N}\mathbf{v}^e)\, dA$. For linear triangular elements:

$$\mathbf{M}^e = \int_{\Omega_e} \mathbf{N}^T\mathbf{N}\, dA = \frac{A_e}{12} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} \tag{22}$$

C. Element Load Vector, $\mathbf{F}^e$ Assuming a constant source $f(x,y) \approx f_e$, the load vector is:

$$\mathbf{F}^e = \int_{\Omega_e} f_e \mathbf{N}^T\, dA = f_e \frac{A_e}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \tag{23}$$

5.4 Finite Element Discretization using Quadrilateral Elements

The domain Ω can alternatively be discretized using four-noded quadrilateral elements. This section develops the element matrices using the **isoparametric formulation**, where the geometry and the unknown field u are interpolated using the same shape functions.

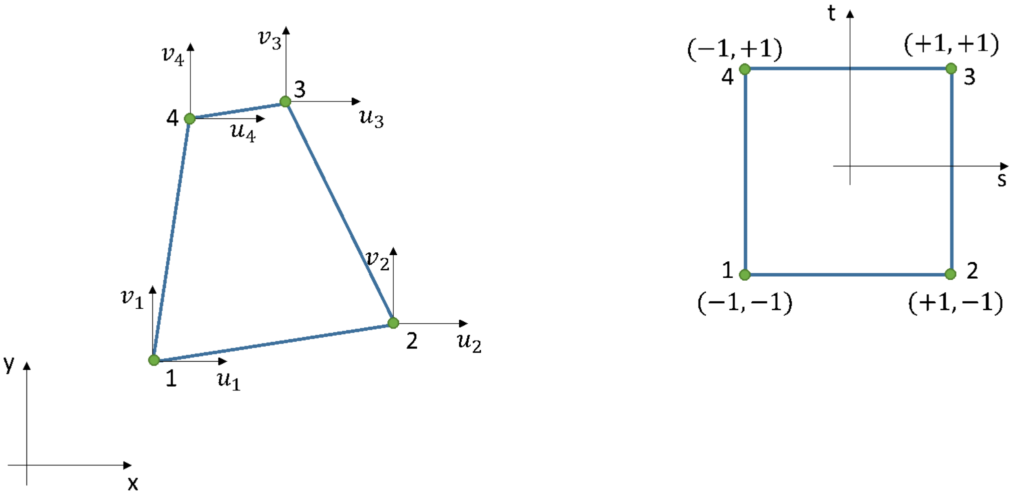

Figure 3: Linear Quadrilateral element

5.4.1 Quadrilateral Shape Functions and Isoparametric Mapping

The shape functions $N_i(s,t)$ for the 4-noded element in the natural coordinate system $(s,t) \in [-1,1]$ are:

$$N_1 = \frac{1}{4}(1-s)(1-t), \qquad N_3 = \frac{1}{4}(1+s)(1+t)$$
$$N_2 = \frac{1}{4}(1+s)(1-t), \qquad N_4 = \frac{1}{4}(1-s)(1+t)$$

The **isoparametric mapping** for the field variable u is:

$$u = \sum_{i=1}^{4} N_i(s,t)u_i = \mathbf{N}(s,t)\mathbf{u}^e \tag{24}$$

5.4.2 Coordinate Transformation and the Jacobian

To calculate the necessary global derivatives, $\partial u/\partial x$ and $\partial u/\partial y$, we use the chain rule relationship defined by the **Jacobian matrix J**:

$$\begin{pmatrix} \partial u/\partial s \\ \partial u/\partial t \end{pmatrix} = \mathbf{J} \begin{pmatrix} \partial u/\partial x \\ \partial u/\partial y \end{pmatrix}, \quad \text{where} \quad \mathbf{J} = \begin{pmatrix} \partial x/\partial s & \partial y/\partial s \\ \partial x/\partial t & \partial y/\partial t \end{pmatrix}$$

The required global derivatives are found by inverting this system:

$$\begin{pmatrix} \partial u/\partial x \\ \partial u/\partial y \end{pmatrix} = \mathbf{J}^{-1} \begin{pmatrix} \partial u/\partial s \\ \partial u/\partial t \end{pmatrix}$$

5.4.3 The B Matrix in Global Coordinates

The derivatives of the shape functions with respect to the natural coordinates are grouped into the $\mathbf{B}_{st}$ matrix:

$$\begin{pmatrix} \partial u/\partial s \\ \partial u/\partial t \end{pmatrix} = \mathbf{B}_{st}\mathbf{u}^e, \quad \text{where} \quad \mathbf{B}_{st} = \begin{pmatrix} \partial N_1/\partial s & \dots & \partial N_4/\partial s \\ \partial N_1/\partial t & \dots & \partial N_4/\partial t \end{pmatrix}$$

The **Gradient Operator Matrix** (**B**) that relates the global gradient ∇u to the nodal values $\mathbf{u}^e$ is thus:

$$\mathbf{B}(s,t) = \mathbf{J}^{-1}(s,t)\mathbf{B}_{st}(s,t) \tag{25}$$

Note that **B** is now a function of s and t.

5.4.4 Element Matrix Integration

The element matrices are obtained by transforming the weak form integral over the physical element area dA to an integral over the canonical square $dsdt$, using the relationship $dA = \det(\mathbf{J})\, dsdt$.

A. Element Stiffness (Conduction) Matrix, $\mathbf{K}^e$

$$\mathbf{K}^e = \int_{-1}^{1}\int_{-1}^{1} \mathbf{B}^T(s,t)\mathbf{B}(s,t)\det(\mathbf{J})\, dsdt$$

B. Element Mass Matrix, $\mathbf{M}^e$

$$\mathbf{M}^e = \int_{-1}^{1}\int_{-1}^{1} \mathbf{N}^T(s,t)\mathbf{N}(s,t)\det(\mathbf{J})\, dsdt$$

C. Element Load Vector, $\mathbf{F}^e$ Assuming a constant source $f(x,y) \approx f_e$:

$$\mathbf{F}^e = f_e \int_{-1}^{1}\int_{-1}^{1} \mathbf{N}^T(s,t)\det(\mathbf{J})\, dsdt$$

These integrals are generally computed numerically using **Gaussian Quadrature**.

5.5 Implementation of Boundary Conditions

The boundary integral term in the weak form, $\oint_{\partial\Omega} v \left(\frac{\partial u}{\partial n}\right)\, d\Gamma$, is the mechanism used to enforce the **natural** and **mixed** boundary conditions. The **essential** (Dirichlet) conditions are handled directly on the nodal variables.

5.5.1 Dirichlet (Essential) Boundary Condition

The Dirichlet condition prescribes the value of the field at a specified boundary node, $u_i = u_0$. Since this is an essential constraint, it must be enforced directly on the algebraic system $\mathbf{Au} = \mathbf{B}$. Common implementation methods include the **elimination method** (or substitution method), the **Lagrange multiplier augmentation**, or the **penalty method**. The elimination method remains the most straightforward for linear systems, involving setting the diagonal matrix entry to one and modifying the load vector to enforce the prescribed value u_0.

5.5.2 Neumann (Natural) and Radiation Boundary Conditions

The natural conditions are incorporated by substituting the known boundary value into the boundary integral term $\mathbf{B}_{\partial\Omega}$:

$$\mathbf{B}_{\partial\Omega} = \oint_{\partial\Omega} v \left(\frac{\partial u}{\partial n} \right) \, d\Gamma \tag{26}$$

The boundary $\partial\Omega$ is a collection of line segments (edges of the elements) where the condition is applied. We discretize this line integral using element shape functions along the boundary edge Γ_e.

Neumann Boundary Condition (Prescribed Flux) If a constant flux $q_0 = \frac{\partial u}{\partial n}$ is prescribed along an element edge Γ_e, the elemental contribution to the load vector is:

$$\mathbf{F}_q^e = \int_{\Gamma_e} v q_0 \, d\Gamma = q_0 \int_{\Gamma_e} \mathbf{N}^T \, d\Gamma \tag{27}$$

A. Application to a 3-Noded Triangular Element Consider a linear triangular element where the boundary condition is applied along the edge connecting nodes i and j, with length L_{ij}. The shape function N_k for the third node is zero along this edge. Since $v = \mathbf{Nv}^e$, the integral simplifies to the contribution from N_i and N_j. The line integral of a linear shape function over its corresponding edge is $L_{ij}/2$. The elemental flux load vector $\mathbf{F}_q^e$ (for a constant flux q_0) is:

$$\mathbf{F}_q^e = q_0 \int_{\Gamma_{ij}} \begin{bmatrix} N_i \\ N_j \\ N_k \end{bmatrix} \, d\Gamma = q_0 \frac{L_{ij}}{2} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \tag{28}$$

The contribution is distributed equally to the two active nodes on the boundary edge.

B. Application to a 4-Noded Quadrilateral Element For a quadrilateral element, the boundary condition is applied along one of the four edges (e.g., the edge between nodes 1 and 2). The integral must be computed using the natural coordinate system ds and the Jacobian of the boundary transformation: $d\Gamma = \frac{L_{ij}}{2} ds$. The elemental flux load vector is:

$$\mathbf{F}_q^e = q_0 \int_{-1}^{1} \mathbf{N}^T(s, t = -1) \left(\frac{dL}{ds} \right) ds \tag{29}$$

For a straight edge (e.g., 1-2), the length scale $\frac{dL}{ds}$ is constant. Since the shape functions along an edge are equivalent to 1D linear shape functions, the resulting vector for a constant flux q_0 on the edge 1-2 is:

$$\mathbf{F}_q^e = q_0 \frac{L_{12}}{2} \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} \tag{30}$$

5.5.3 Absorbing) Boundary Condition (ABC)

The first-order Sommerfeld radiation condition, $\frac{\partial u}{\partial n} + iku = 0$ on the boundary Γ_{Som}, is a **Robin (mixed) BC**. Substituting the relationship $\frac{\partial u}{\partial n} = -iku$ into the weak form boundary term $\mathbf{B}_{\partial\Omega}$ (Equation (26)) yields:

$$\oint_{\Gamma_{\text{Som}}} v \left(\frac{\partial u}{\partial n} \right) d\Gamma = \oint_{\Gamma_{\text{Som}}} v(-iku)\, d\Gamma$$

This term must be moved to the left-hand side of the global system equation, contributing to the system matrix **A**. Recalling $u = \mathbf{N}\mathbf{u}^e$ and $v = \mathbf{N}\mathbf{v}^e$, this contribution yields the element **boundary impedance matrix $\mathbf{K}_I^e$**:

$$\mathbf{K}_I^e = ik \int_{\Gamma_e} \mathbf{N}^T \mathbf{N}\, d\Gamma \tag{31}$$

The term $\int_{\Gamma_e} \mathbf{N}^T \mathbf{N}\, d\Gamma$ is equivalent to the 1D element mass matrix integrated along the boundary edge (line integral). This matrix is assembled into the global system matrix **A** corresponding to the boundary nodes.

A. Application to a 3-Noded Triangular Element Consider a linear triangular element where the ABC is applied along the edge Γ_{ij} connecting nodes i and j, with length L_{ij}. The shape function N_k for the third node is zero along this edge. The resulting 3×3 matrix contribution is:

$$\mathbf{K}_I^e = ik \int_{\Gamma_{ij}} \begin{pmatrix} N_i \\ N_j \\ N_k \end{pmatrix} \begin{pmatrix} N_i & N_j & N_k \end{pmatrix} d\Gamma = ik \frac{L_{ij}}{6} \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{32}$$

B. Application to a 4-Noded Quadrilateral Element For the 4-noded quadrilateral element, the ABC is applied along one edge Γ_e. For a linear edge of length L, the shape functions along that edge reduce to the 1D linear case. For example, if the ABC is on the edge connecting nodes 1 and 2 (length L_{12}):

$$\mathbf{K}_I^e = ik\frac{L_{12}}{6}\begin{pmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \tag{33}$$

This term is added to the corresponding entries in the global system matrix **A**.

6 Implementation: 2D Helmholtz Solver Architecture

This section details the development and architecture of the 2D Helmholtz equation solver, which utilizes a hybrid programming approach for performance and usability. The solver is built using C#, OpenTK, and C++, combining a modern graphical interface with a high-performance numerical core.

6.1 Software Components

The entire system is divided into two primary components:

1. **Front-End (GUI and Visualization):** Developed using **C#** and the **OpenTK** library. The C# layer provides a robust Graphical User Interface (GUI) that handles:
   - Importing 2D finite element meshes (e.g., triangular or quadrilateral elements).
   - Defining and applying all boundary conditions (Dirichlet, Neumann, Sommerfeld).
   - Initiating the simulation and providing various visualization options for the resultant field $u(x, y)$.
2. **Back-End (Solver Core):** Developed in high-performance **C++**. This component implements the finite element methods derived in the previous sections, specifically handling:
   - Global matrix assembly (**Au** $=$ **B**).
   - Applying complex boundary conditions (Sommerfeld ABC).
   - Solving the resulting large, sparse linear system of equations.

6.2 Interoperability and Performance

The integration between the C# front-end and the C++ back-end is achieved using **P/Invoke** (Platform Invoke) or a similar interoperability layer. This approach ensures that performance-critical matrix operations remain fast in C++, while benefiting from the rapid development and rich GUI features of the C# environment.

The entire software project is available as open-source code for public access and verification at the repository: `https://github.com/Samson-Mano/`

`2DHelmholtz_solver`. The following sections demonstrate various problems solved and validated using this tool.

7 Numerical notes and pitfalls

- The Helmholtz operator is indefinite for real k; for large k the linear system can be harder to solve (special solvers / preconditioners needed in large 2D/3D problems).
- Be aware of resonance: if the domain and k produce standing wave modes that satisfy homogeneous BCs, the continuous operator may have nontrivial nullspace — meshes and BCs need careful handling.
- For open-domain scattering problems use absorbing boundary conditions or perfectly matched layers (PML).
- Increase element order or refine the mesh for accuracy; a rule-of-thumb is several elements per wavelength ($\lambda = 2\pi/k$), e.g. 8–10 elements per wavelength for linear elements.

8 Conclusion and next steps

This document has the derivation, FEM weak form, element matrices and a minimal solver. Next steps you may want:

1. Run the Python implemention and compare with the analytic solution in §6.
2. Add plots (Matplotlib) to visualize real/imag parts and error vs mesh refinement.
3. Extend to higher order elements or to 2D.
4. Add absorbing BCs or PML for scattering examples.

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